Exp No:6 Experiment to understand feature scaling.

CODE

```
import numpy as np
import pandas as pd
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import MinMaxScaler
df=pd.DataFrame({
'Country':['USA','France','USA','China','France','USA','China',np.nan,'France','USA'],
    'Age':[22,34,27,np.nan,45,38,31,50,29,41],
'Salary':[35000,60000,48000,72000,np.nan,55000,68000,80000,52000,64000],
    'Purchased':['No','Yes','No','Yes','No','Yes','No','Yes','No','Yes']
})
print("Original Dataset:")
print(df)
print("\n\nDataFrame Information:")
print(df.info())
df['Country']=df['Country'].fillna(df['Country'].mode()[0])
features=df.iloc[:,:-1].values
label=df.iloc[:,-1].values
```

```
age=SimpleImputer(strategy="mean",missing_values=np.nan)
Salary=SimpleImputer(strategy="mean",missing_values=np.nan)
age.fit(features[:,[1]])
Salary.fit(features[:,[2]])
features[:,[1]]=age.transform(features[:,[1]])
features[:,[2]]=Salary.transform(features[:,[2]])
print("\n\nAfter Filling Missing Values:")
print(features)
oh=OneHotEncoder(sparse_output=False)
Country=oh.fit_transform(features[:,[0]])
print("\n\nOne Hot Encoding:")
print(Country)
final_set=np.concatenate((Country,features[:,[1,2]]),axis=1)
print("\n\nFinal Set:")
print(final_set)
sc=StandardScaler()
sc.fit(final_set)
feat_standard_scaler=sc.transform(final_set)
print("\n\nStandard Scaler:")
print(feat_standard_scaler)
mms=MinMaxScaler(feature_range=(0,1))
mms.fit(final_set)
feat_minmax_scaler=mms.transform(final_set)
```

```
print("\n\nMin Max Scaler:")
print(feat_minmax_scaler)
```

OUTPUT

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	NaN	50.0	83000.0	No
9	France	37.0	67000.0	Yes

```
array([['France', 44.0, 72000.0],
       ['Spain', 27.0, 48000.0],
       ['Germany', 30.0, 54000.0],
       ['Spain', 38.0, 61000.0],
       ['Germany', 40.0, 63777.77777777778],
       ['France', 35.0, 58000.0],
       ['Spain', 38.77777777777778, 52000.0],
       ['France', 48.0, 79000.0],
       ['France', 50.0, 83000.0],
       ['France', 37.0, 67000.0]], dtype=object)
```

```
array([[1.0, 0.0, 0.0, 44.0, 72000.0],
       [0.0, 0.0, 1.0, 27.0, 48000.0],
       [0.0, 1.0, 0.0, 30.0, 54000.0],
       [0.0, 0.0, 1.0, 38.0, 61000.0],
       [0.0, 1.0, 0.0, 40.0, 63777.77777777778],
       [1.0, 0.0, 0.0, 35.0, 58000.0],
       [0.0, 0.0, 1.0, 38.77777777777778, 52000.0],
       [1.0, 0.0, 0.0, 48.0, 79000.0],
       [1.0, 0.0, 0.0, 50.0, 83000.0],
       [1.0, 0.0, 0.0, 37.0, 67000.0]], dtype=object)
```

```
array([[1.        , 0.        , 0.        , 0.73913043, 0.68571429],
       [0.        , 0.        , 1.        , 0.        , 0.        ],
       [0.        , 1.        , 0.        , 0.13043478, 0.17142857],
       [0.        , 0.        , 1.        , 0.47826087, 0.37142857],
       [0.        , 1.        , 0.        , 0.56521739, 0.45079365],
       [1.        , 0.        , 0.        , 0.34782609, 0.28571429],
       [0.        , 0.        , 1.        , 0.51207729, 0.11428571],
       [1.        , 0.        , 0.        , 0.91304348, 0.88571429],
       [1.        , 0.        , 0.        , 1.        , 1.        ],
       [1.        , 0.        , 0.        , 0.43478261, 0.54285714]])
```

RESULT Hence the feature scaling has been studied successfully.